



Clinical correlation “Vascular disorder”

รายวิชาระบบหัวใจและหลอดเลือด

29-10-2025

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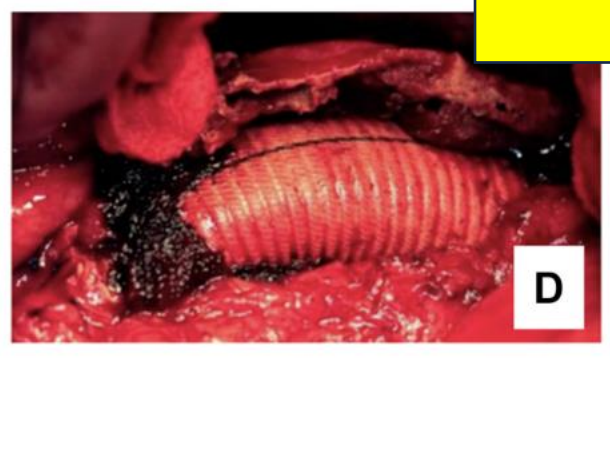
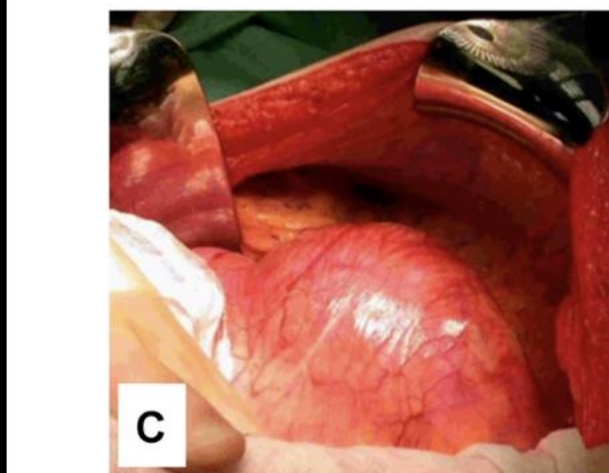
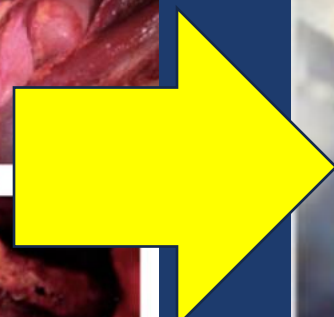
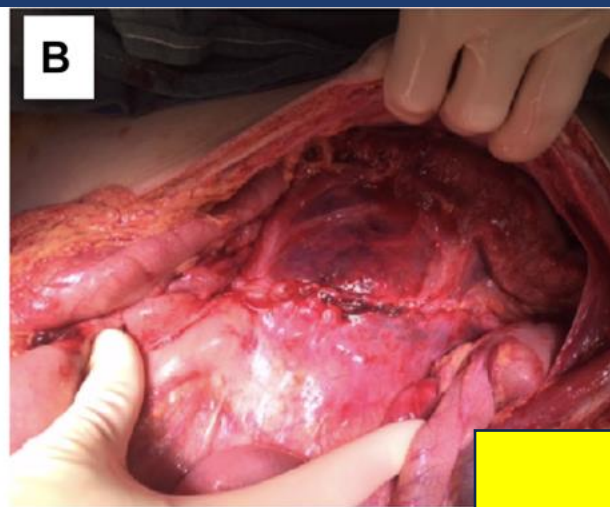
Sorracha Rookapan, MD



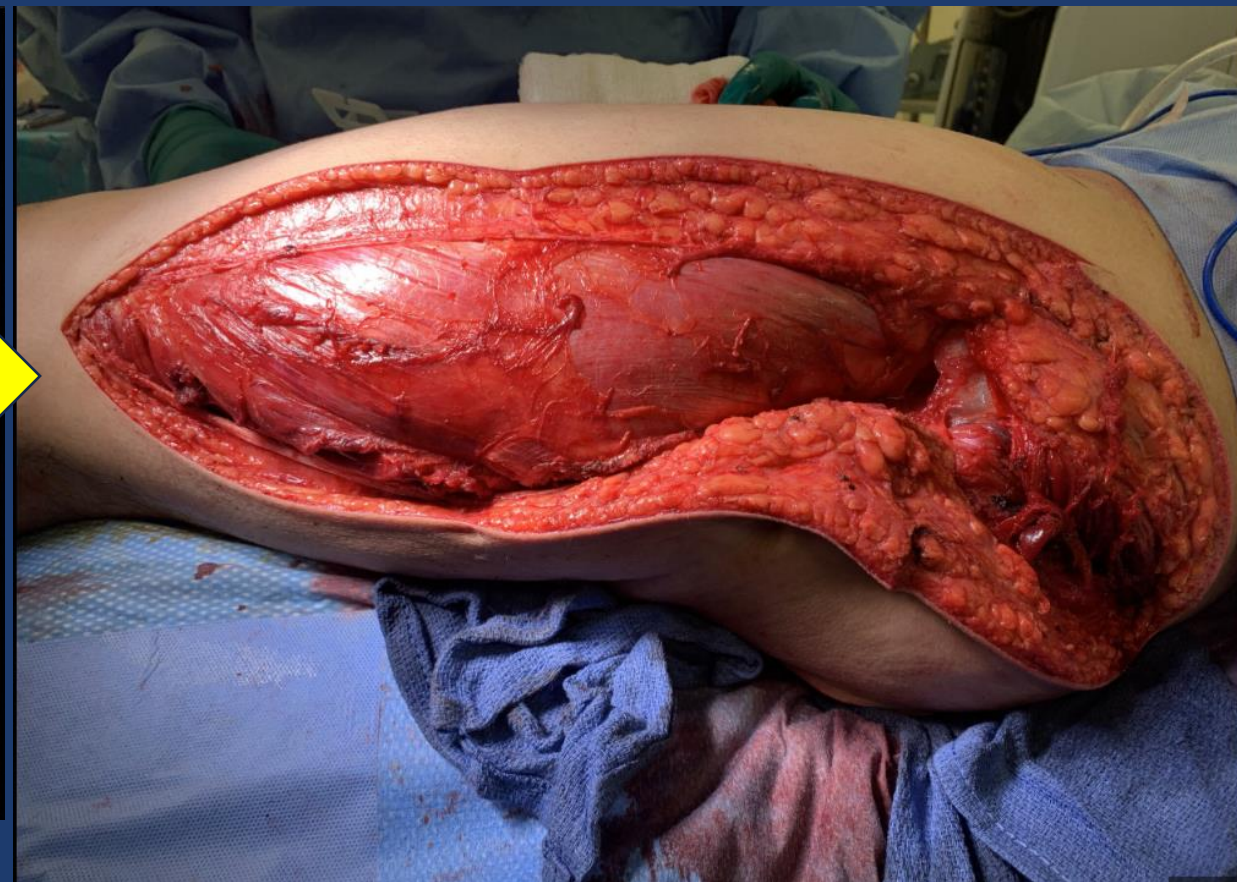
Introduction

How is importance of clinical correlation in vascular patients?

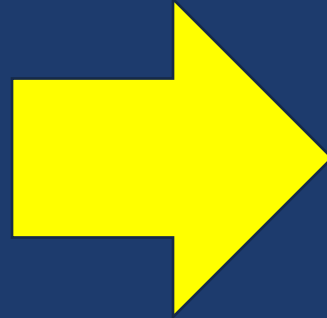
● ruptured AAA



● Acute limb ischemia



Acute DVT



 ● Vascular disorder need specific care!!!

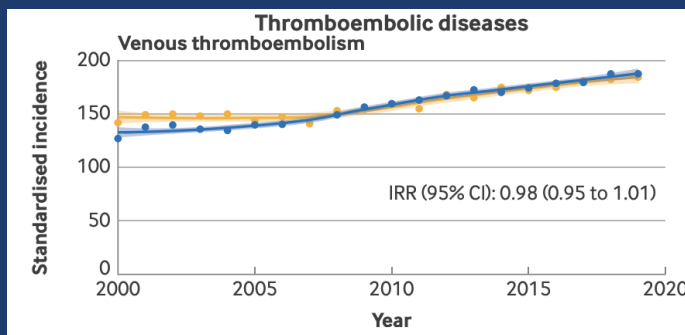
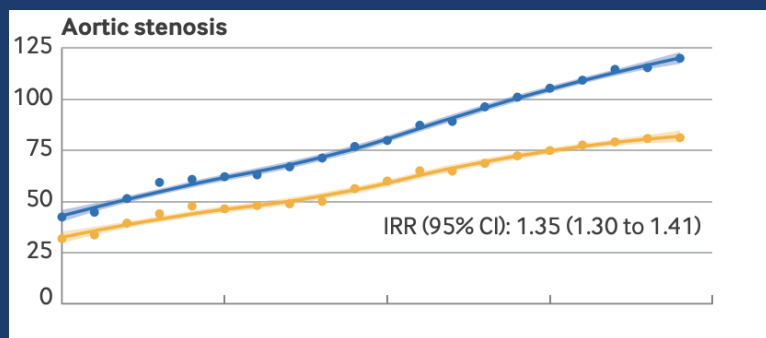
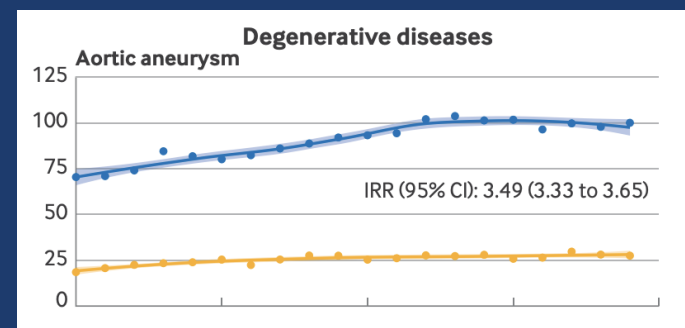
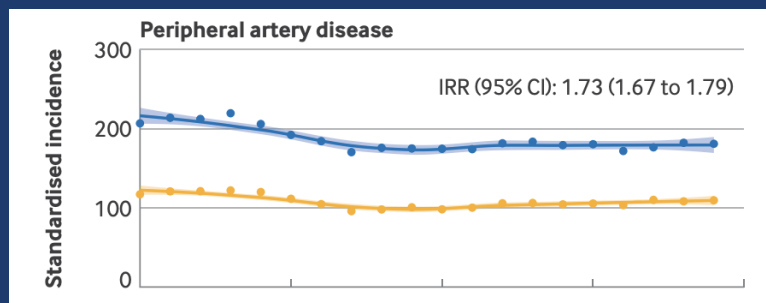
To save life!

To save limb!

Return to normal life activity!

Trends in cardiovascular disease incidence among 22 million people in the UK over 20 years: population based study

Nathalie Conrad,^{1,2,3} Geert Molenberghs,⁴ Geert Verbeke,⁴ Francesco Zaccardi,⁵ Claire Lawson,⁵
 Jocelyn M Friday,¹ Huimin Su,² Pardeep S Jhund,¹ Naveed Sattar,⁶ Kazem Rahimi,³ John G Cleland,¹
 Kamlesh Khunti,⁵ Werner Budts,^{1,7} John J V McMurray¹





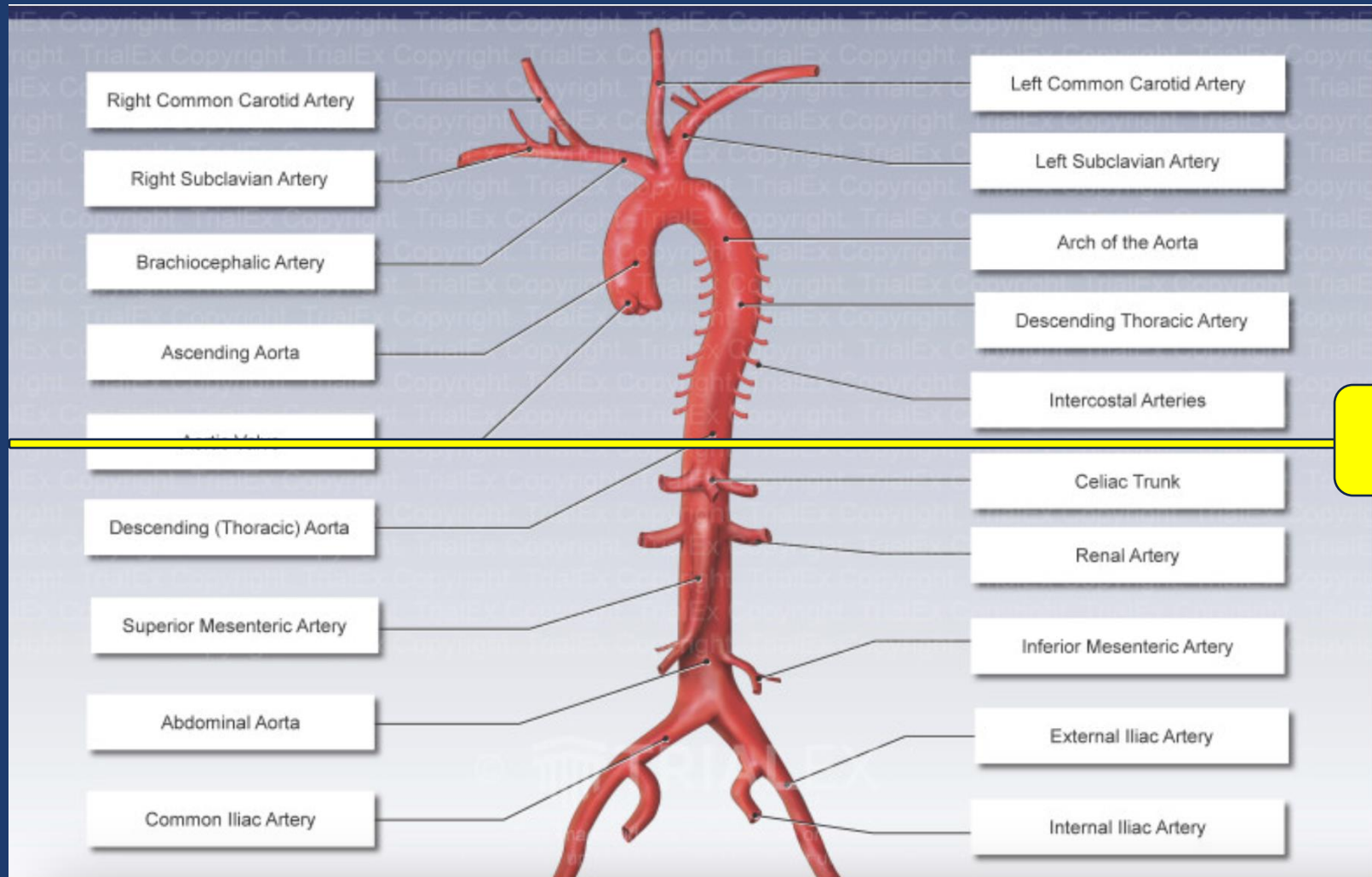
Scope

- ❖ Anatomy and physiology of vascular system
- ❖ Clinical correlation of vascular disorder
- ❖ Pharmacological management of vascular conditions



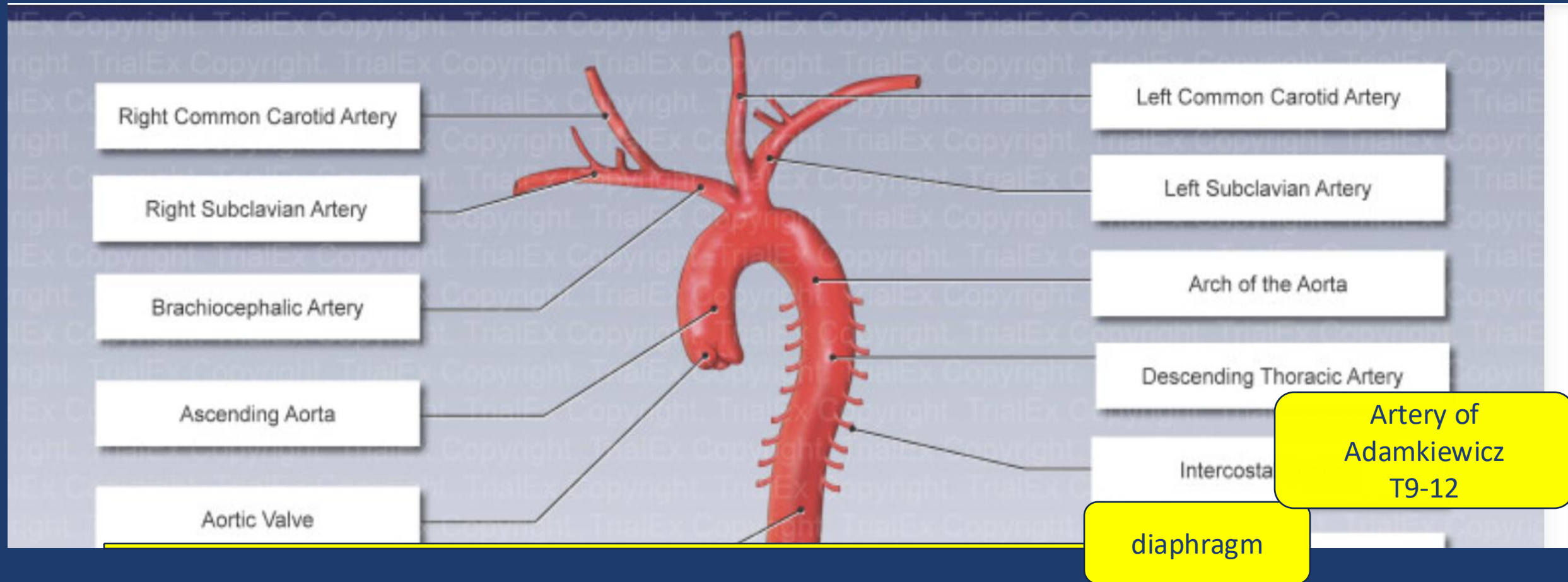
Arterial system

Central; aorta

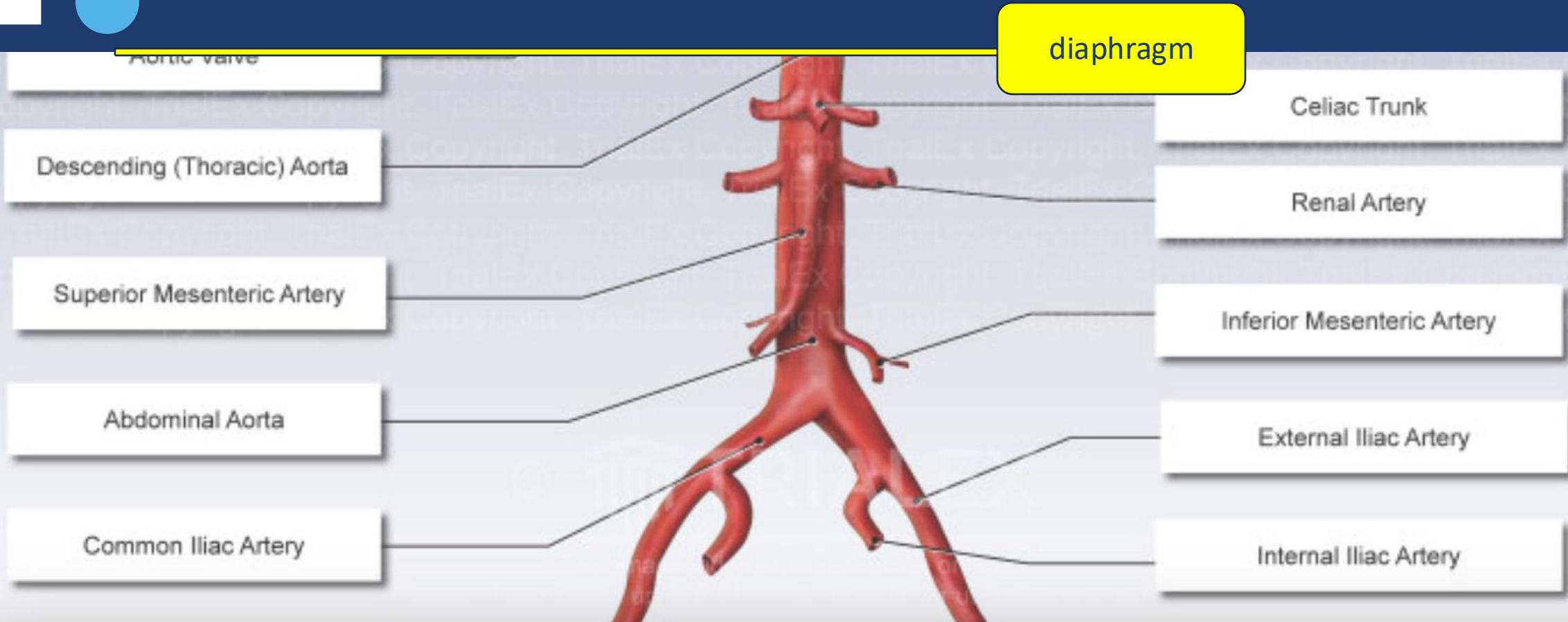


diaphragm

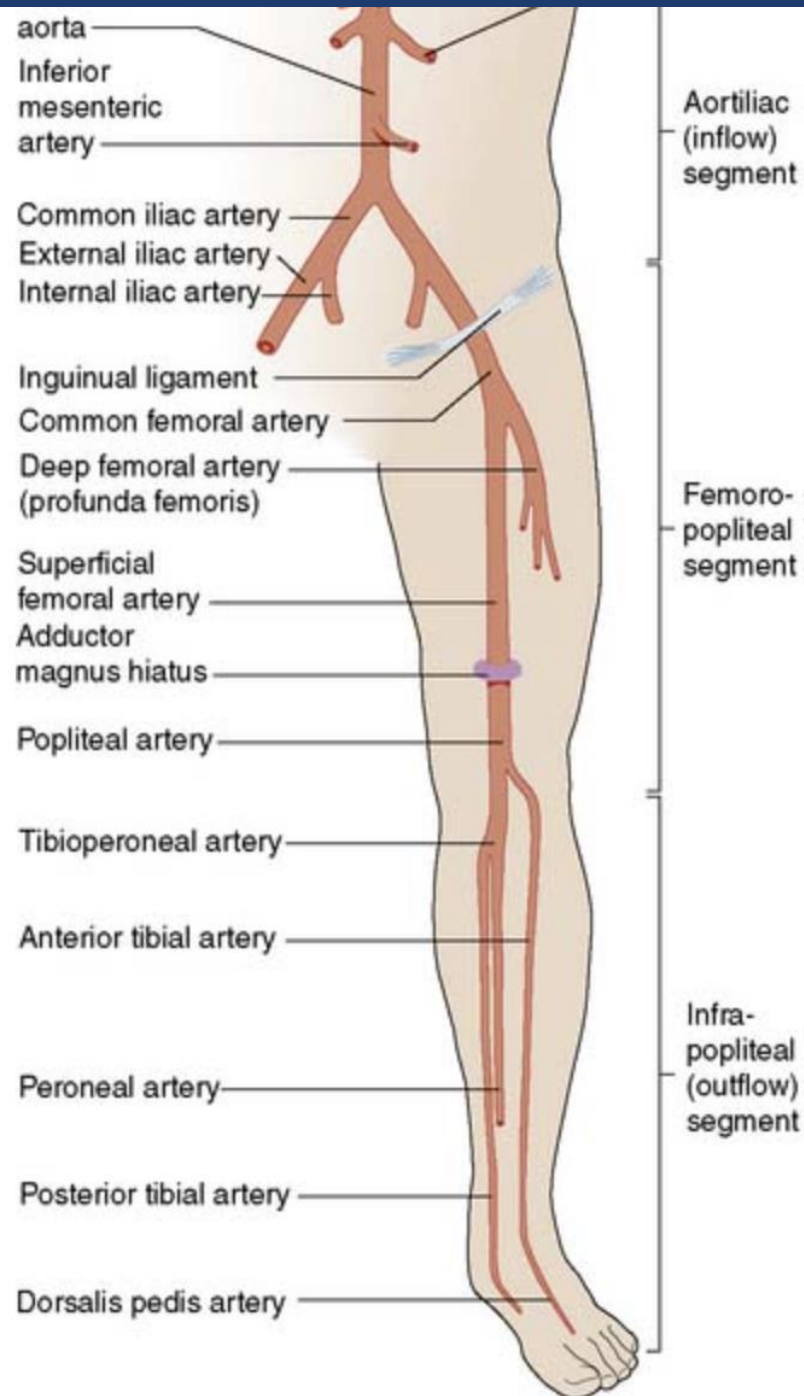
Thoracic aorta



Abdominal aorta

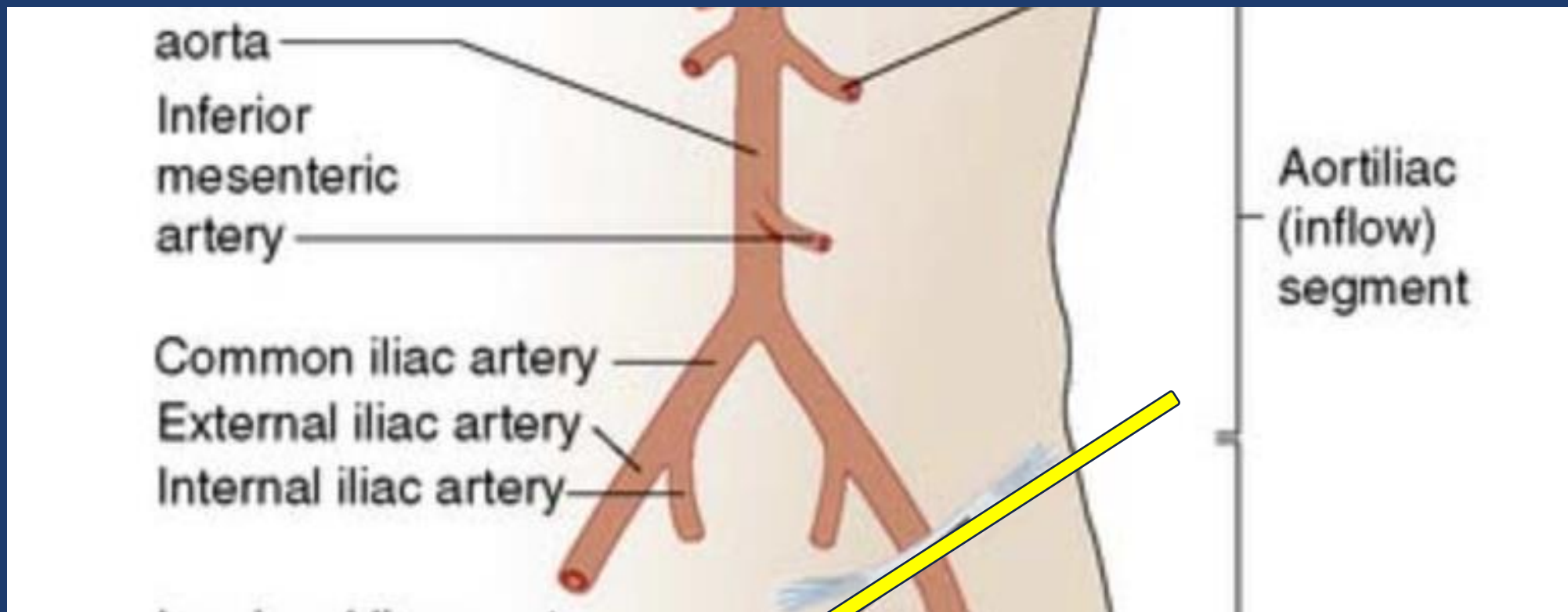


Peripheral artery



Peripheral artery

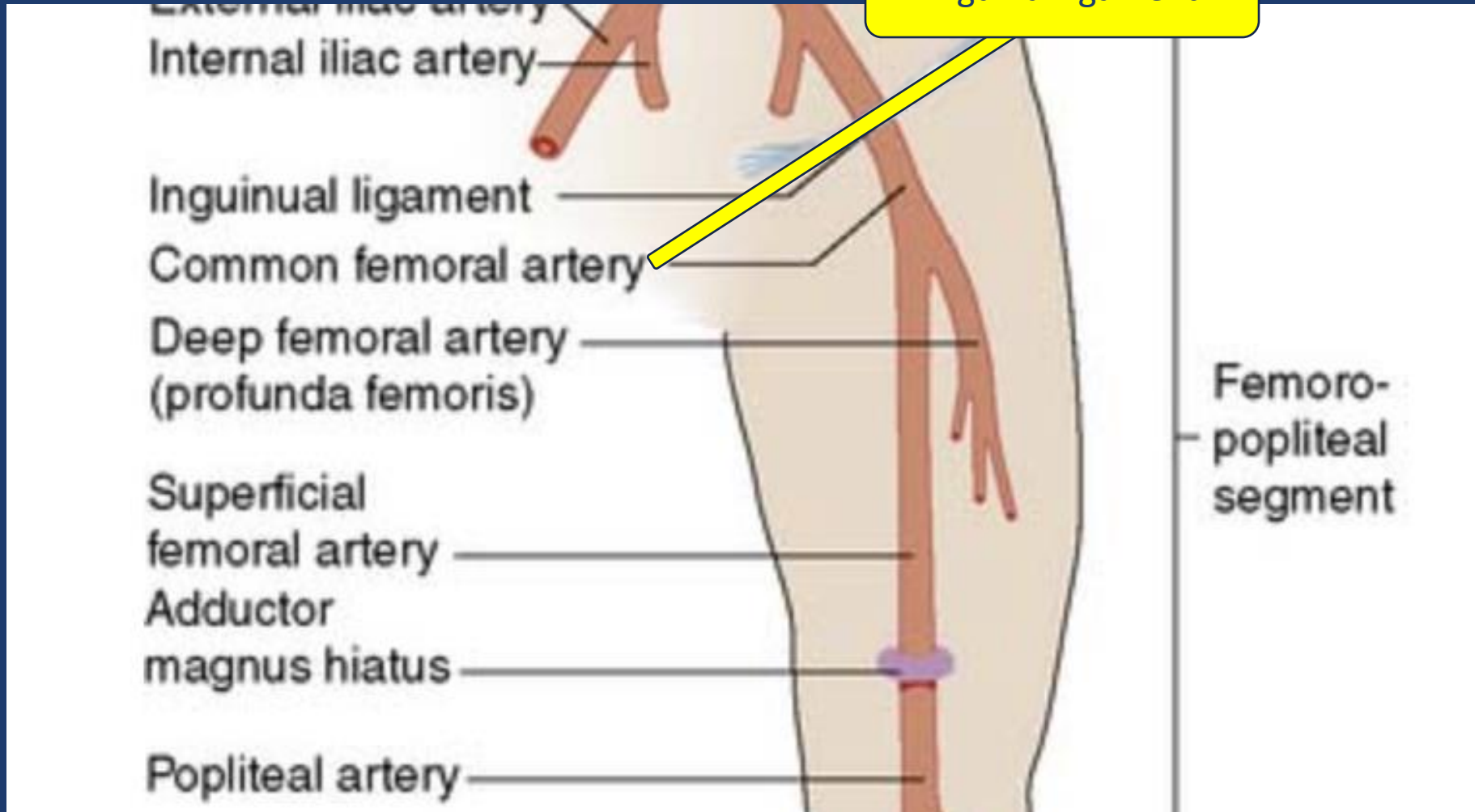
Aorto-iliac segment



Inguinal ligament

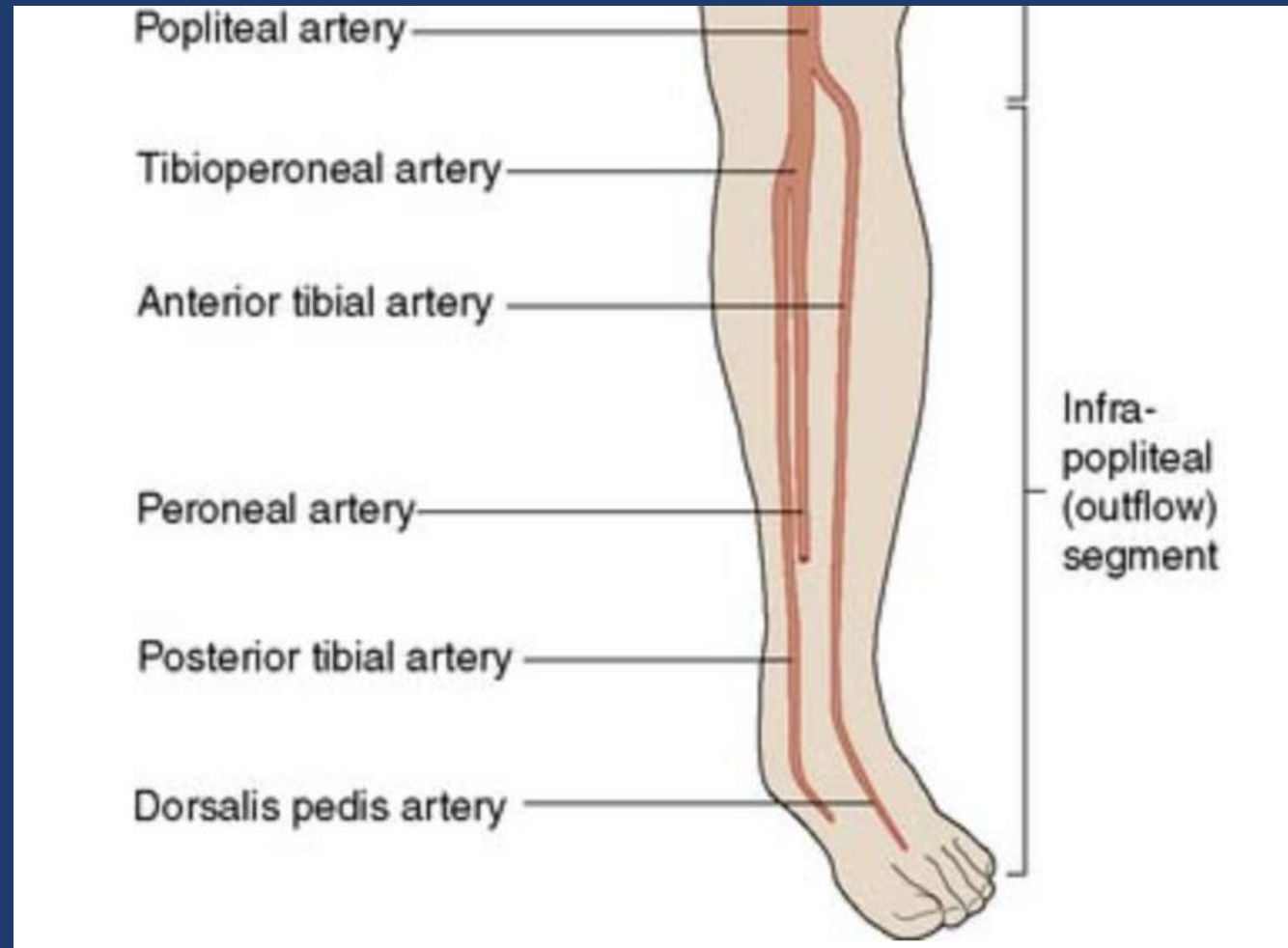
Peripheral artery

Femoro-popliteal segment

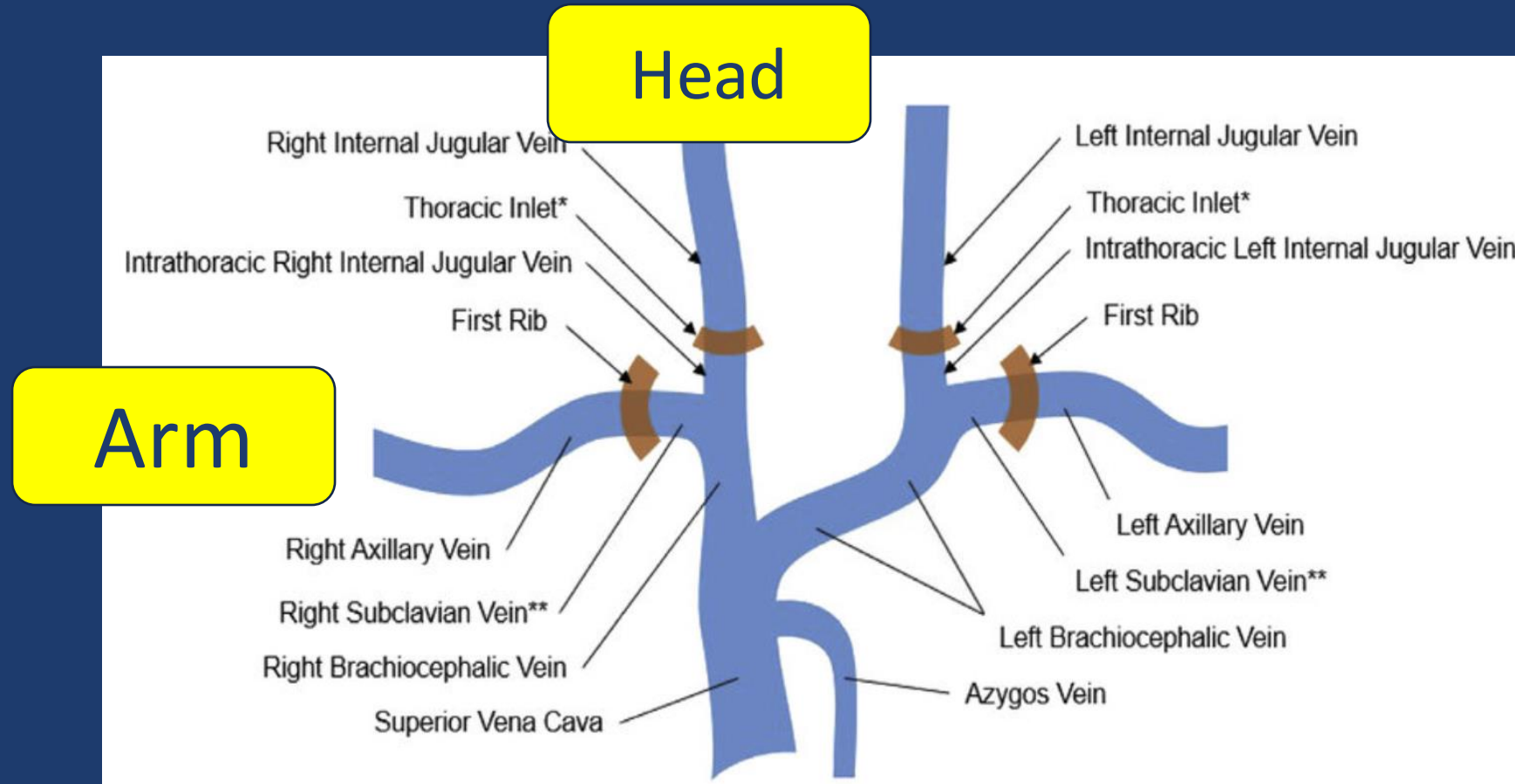


Peripheral artery

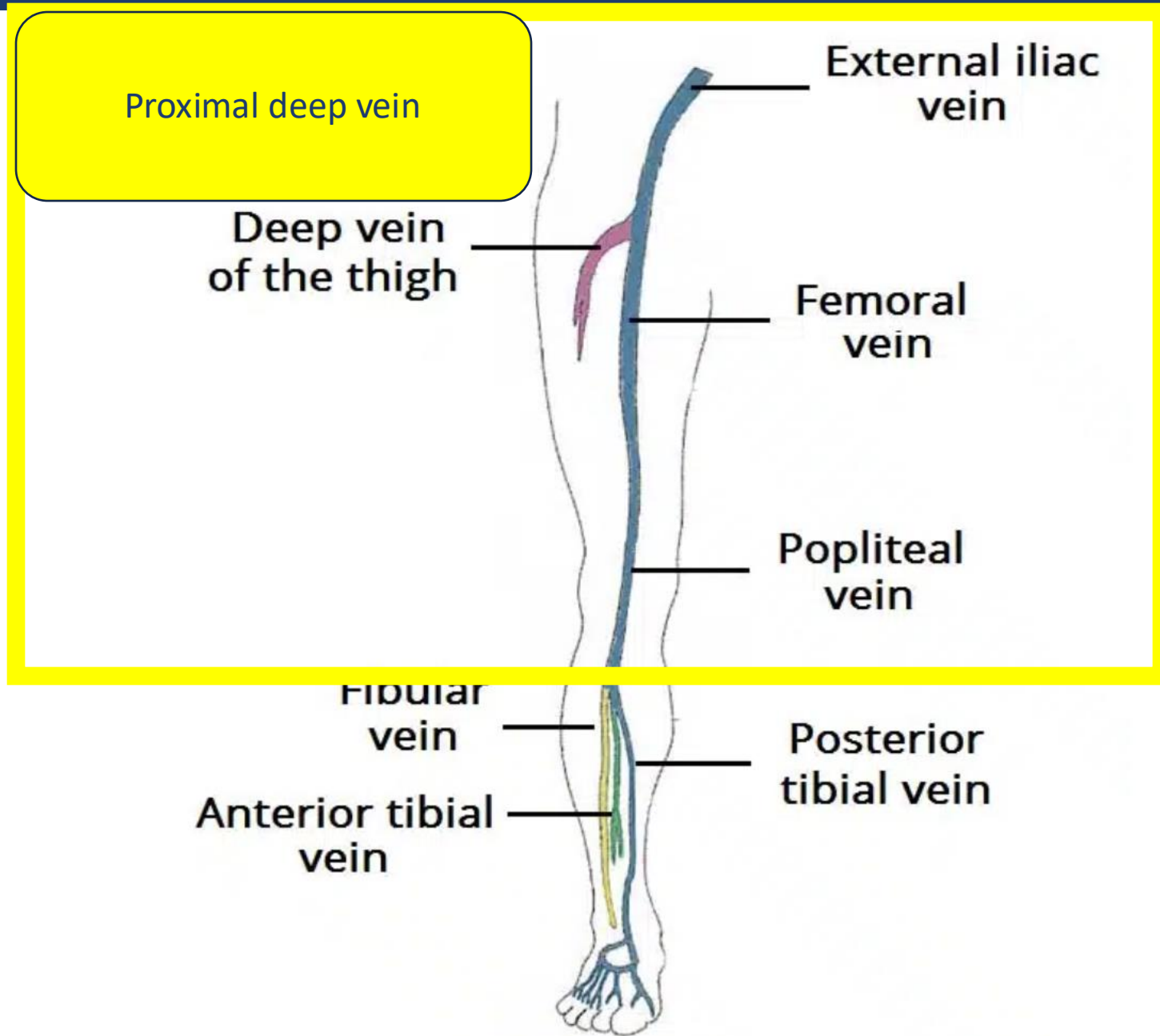
Below the knee segment



● Venous anatomy; Central vein



Venous anatomy



● Venous anatomy

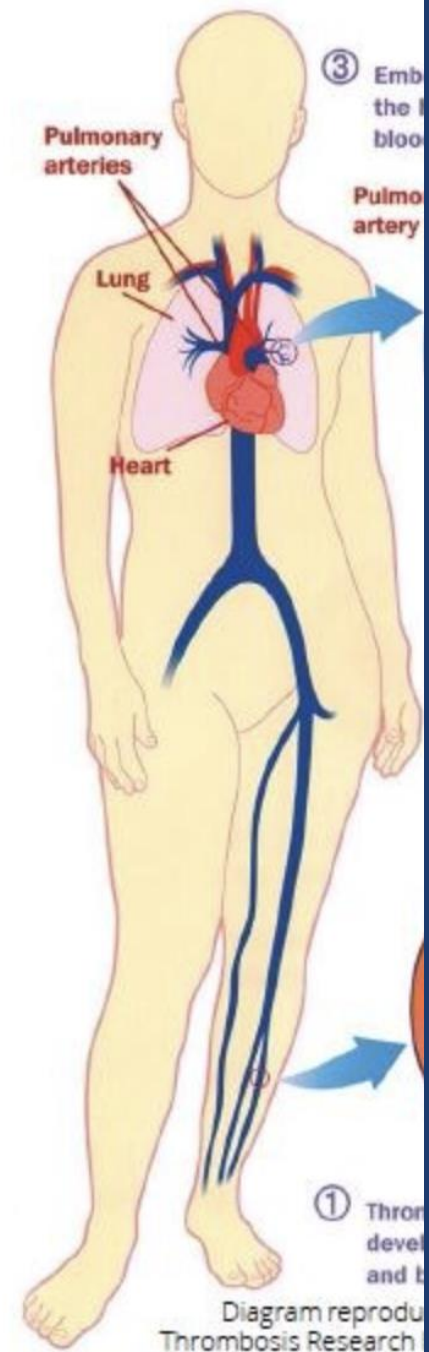
Proximal deep vein

Deep vein
of the thigh

External iliac
vein

Femoral
vein

Popliteal
vein



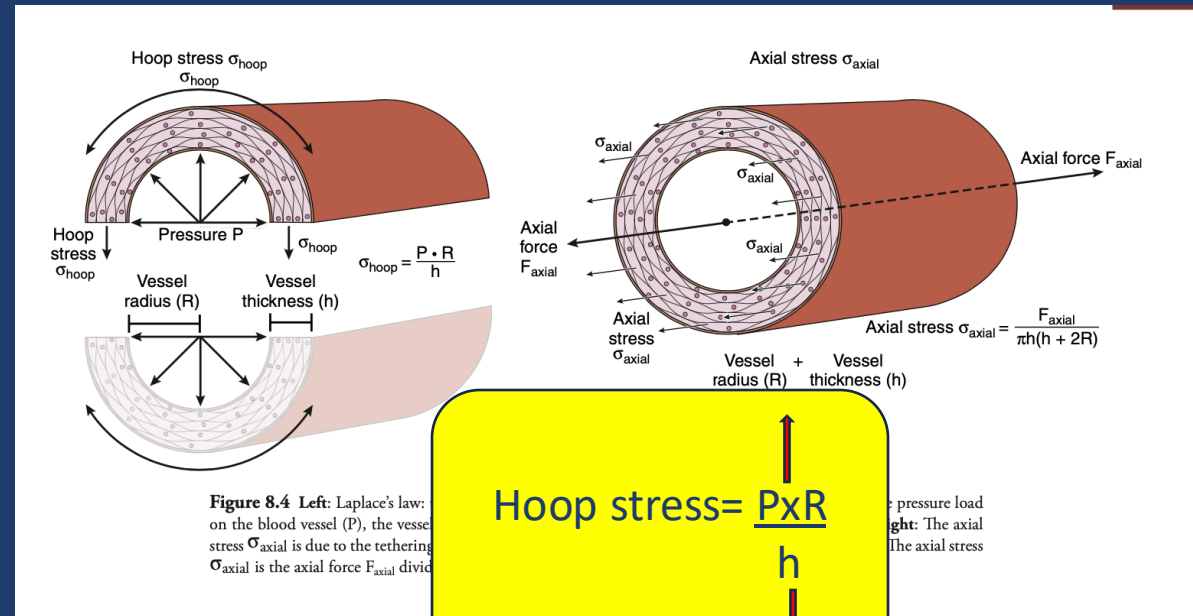
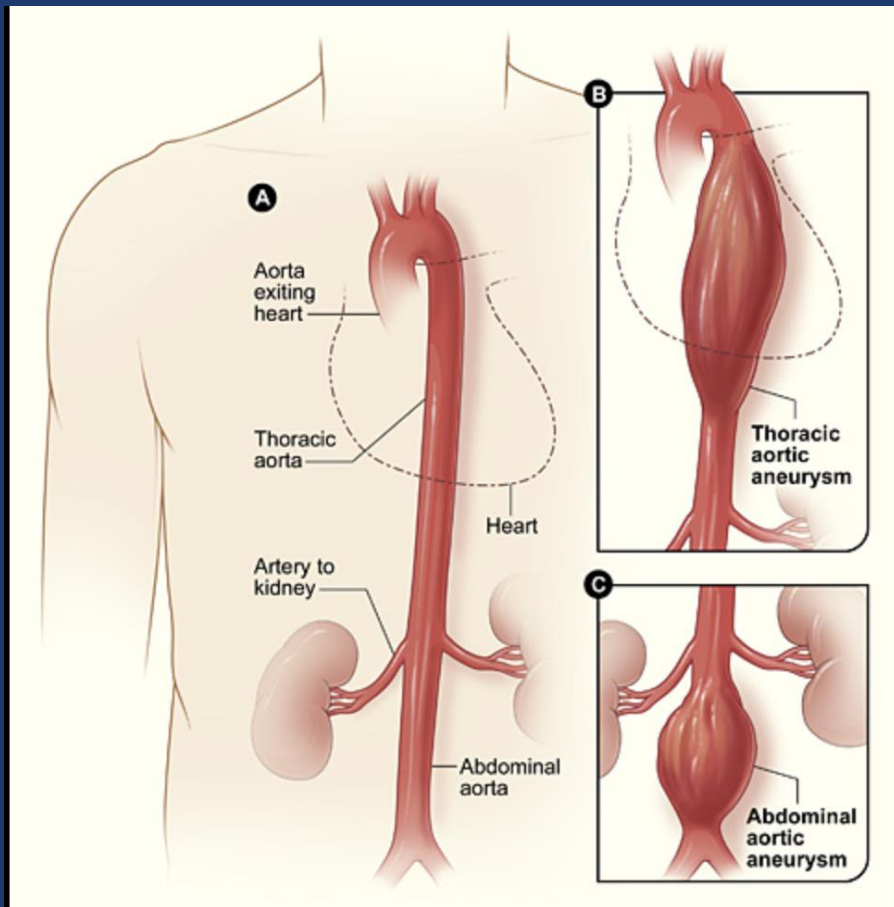


Clinical correlation in vascular disorder

Arterial system; Central

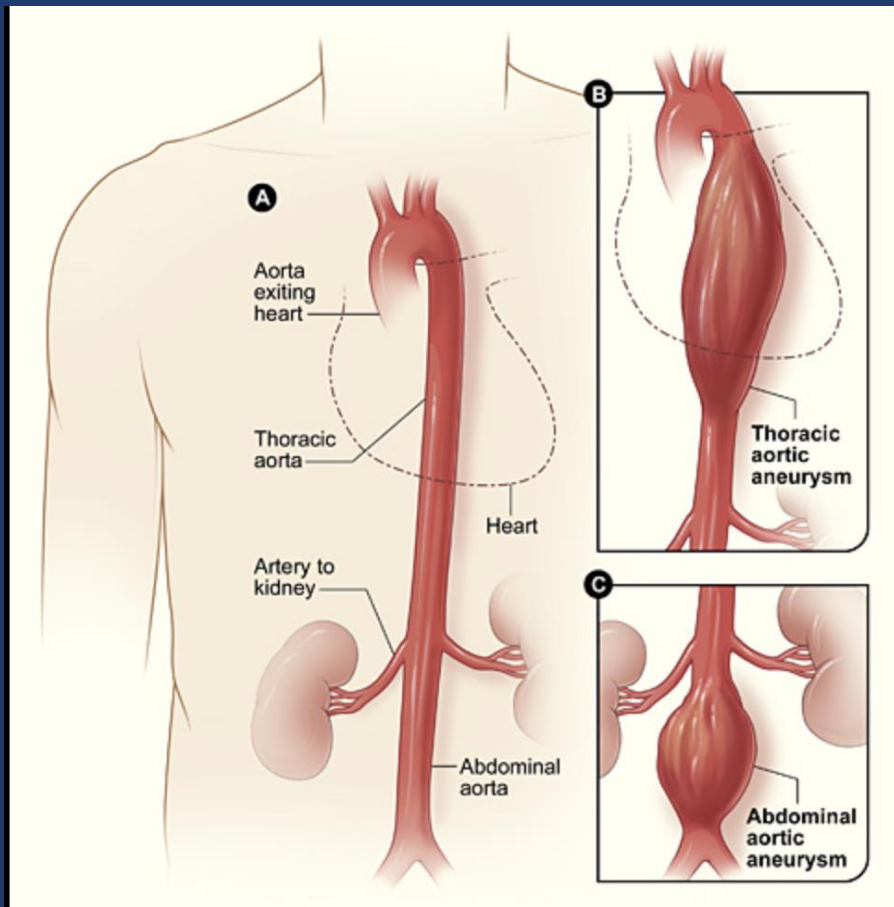
● aortic aneurysm

Definition = 1.5x of normal diameter

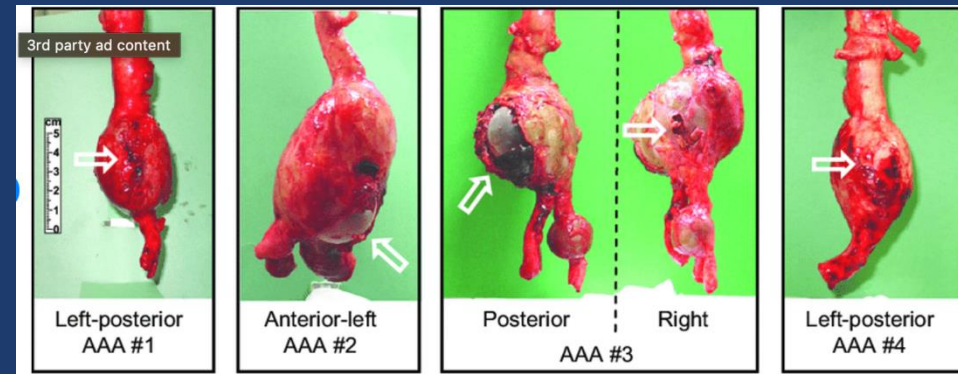


● aortic aneurysm

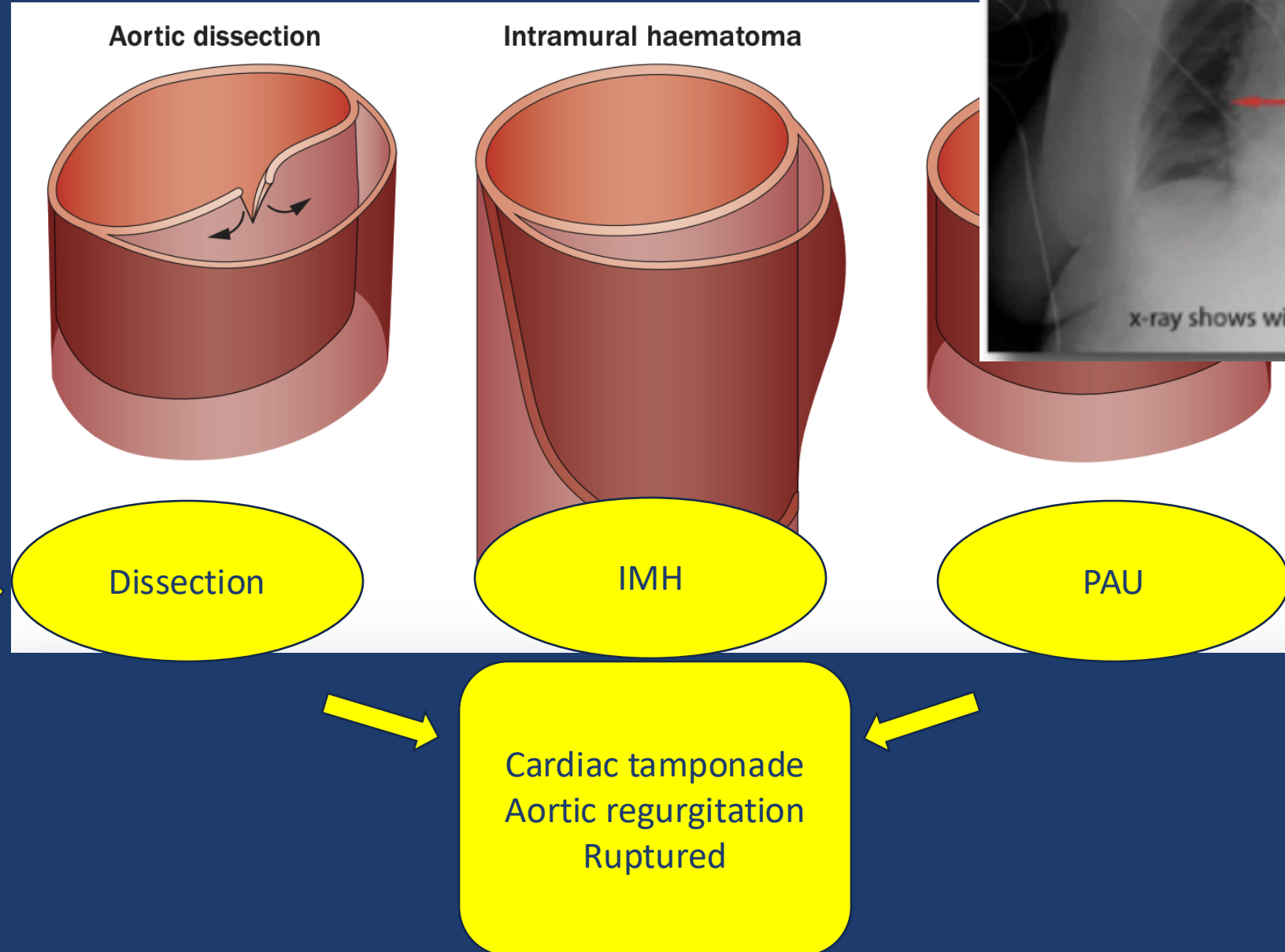
Definition = 1.5x of normal diameter



Ruptured



Acute aortic syndrome





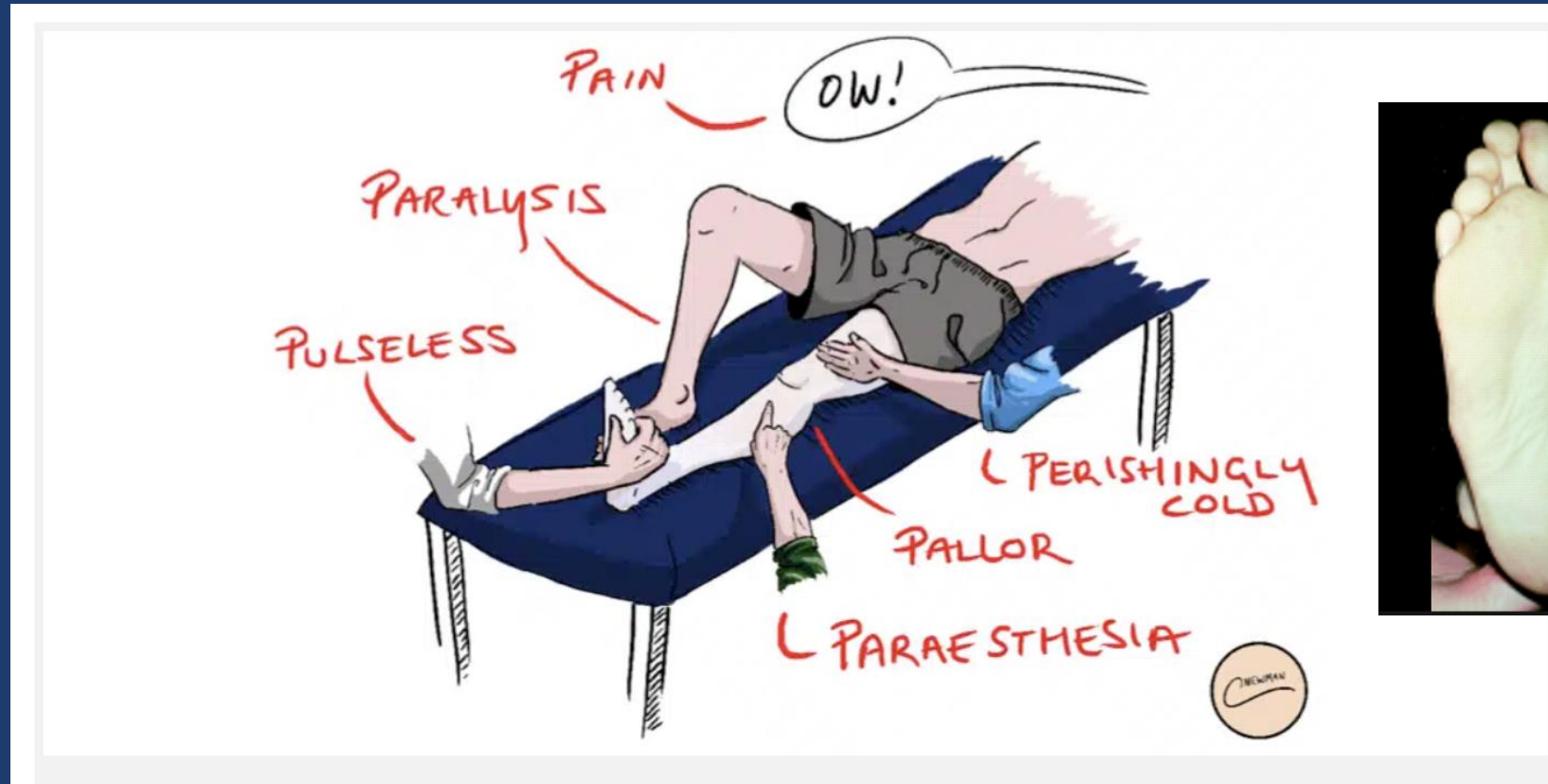
Clinical correlation in vascular disorder

Arterial system; Peripheral

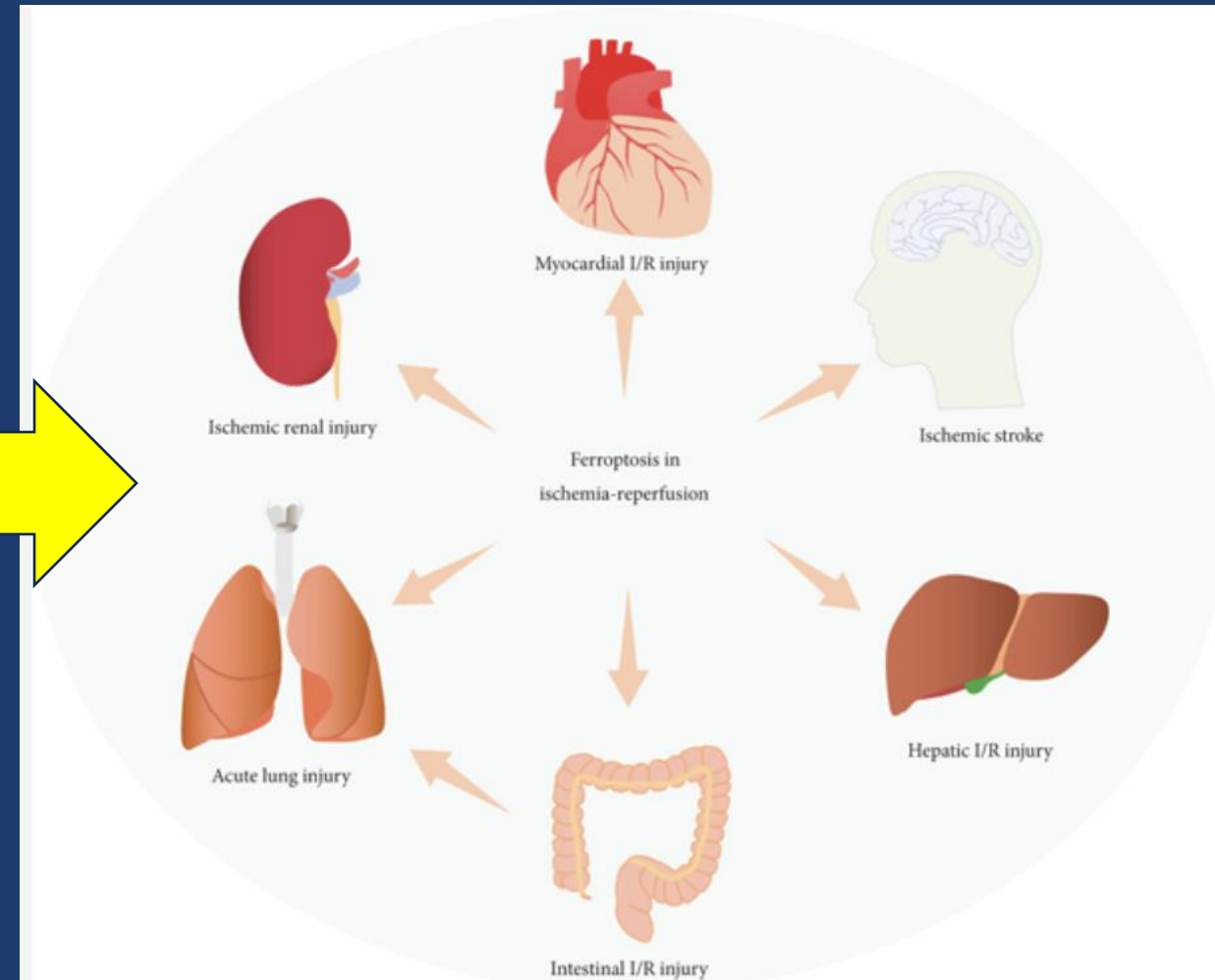
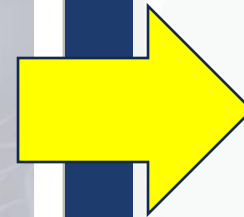
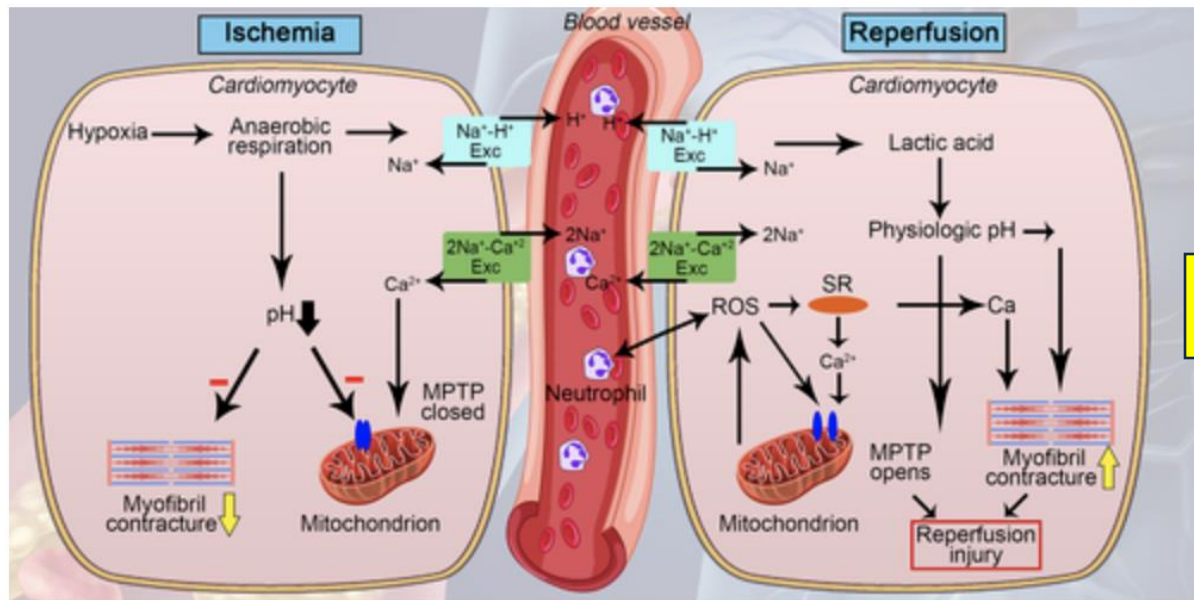
● Acute limb ischemia

6Ps

- Pain
- Pallor
- Pulselessness
- Poikilothermic
- Paranesthesia
- paralysis



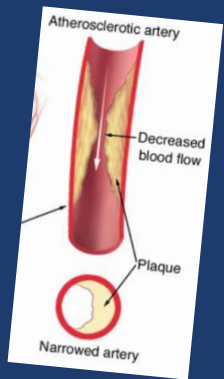
Ischemic-reperfusion injury



Chronic limb threatening ischemia

Peripheral arterial disease + at least one clinical + timing > 2 wk

- Gangrene
- Ulcer
- Rest pain

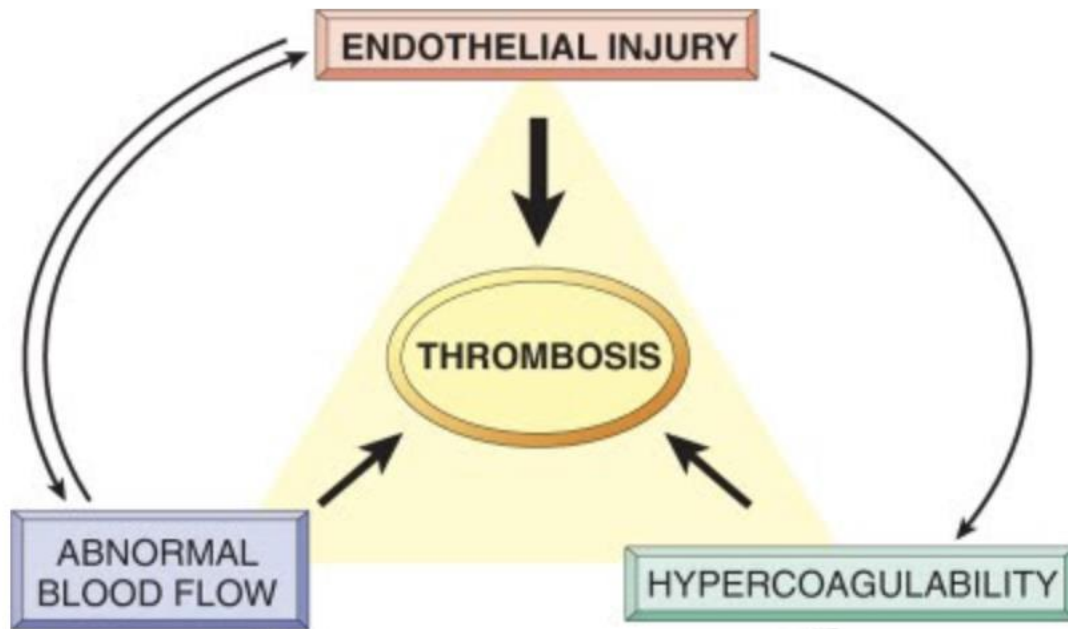




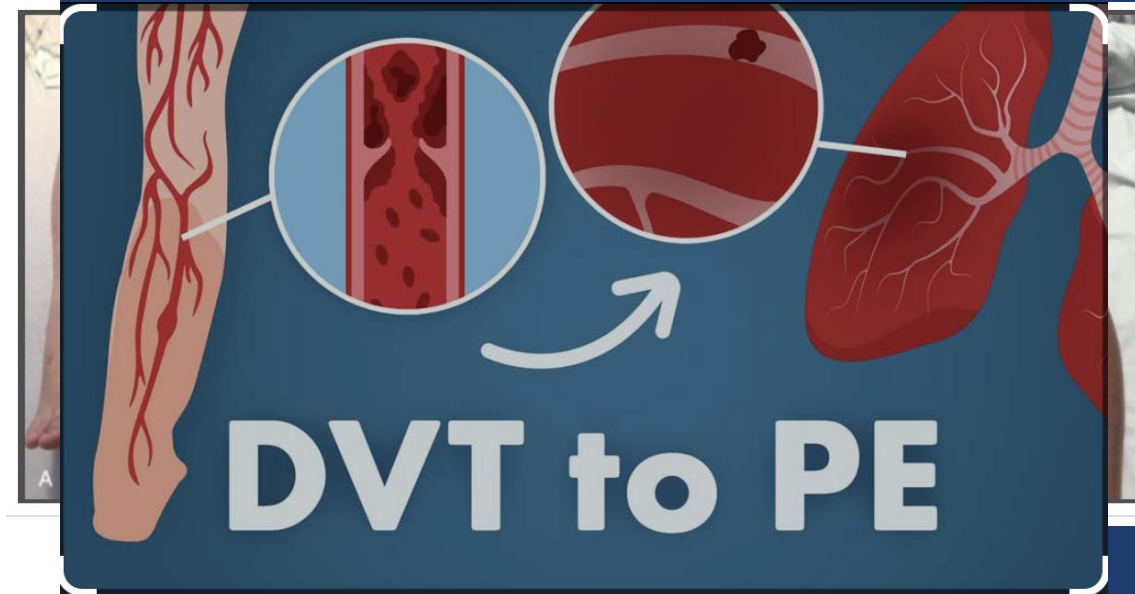
Clinical correlation in vascular disorder

Venous system

● Acute deep vein thrombosis



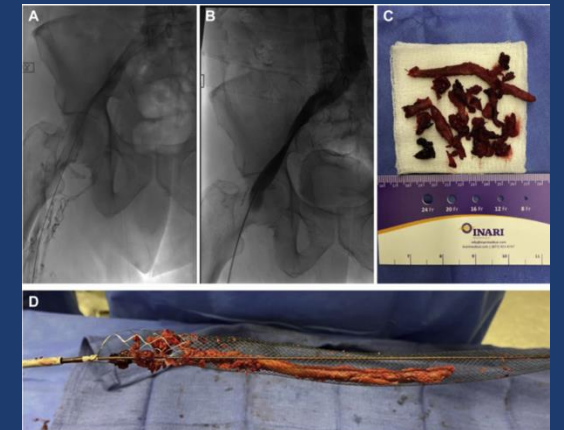
Virchow's triad



● Phlegmasia cerulea dolens

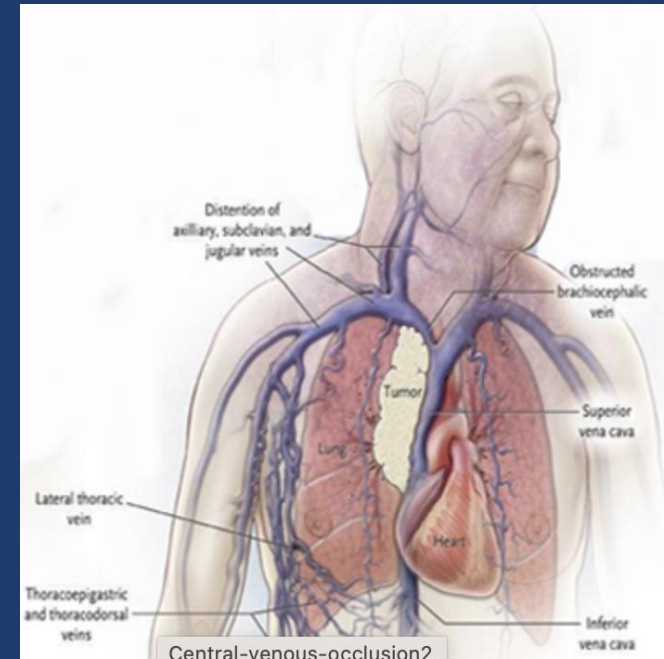


- ❖ Impending venous gangrene
- ❖ High risk to limb loss



Central vein steno-occlusive disease

- Central vein stenosis in hemodialysis patient
- External compression; SVC syndrome



Chronic venous insufficiency



Venous hypertension

C1

C2

C3

C4
(a,b,c)

C5

C6

ge

Chronic inflammation

Chronic venous leg ulcer



Pharmacological Management of Vascular disorder

- ❖ Inotropic

- ❖ Antihypertensive drug

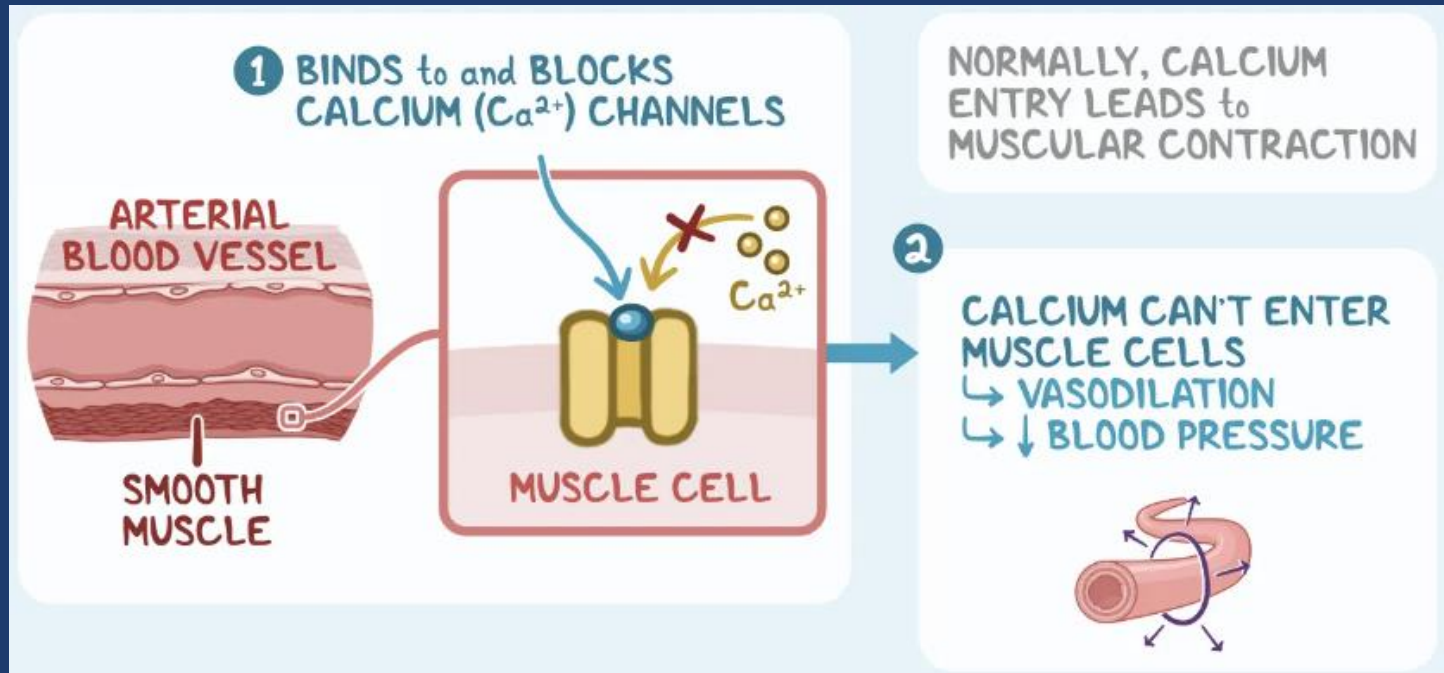
- Nicardipine
- Labetalol

- ❖ rtPA

- ❖ Anticoagulant; unfractionated heparin

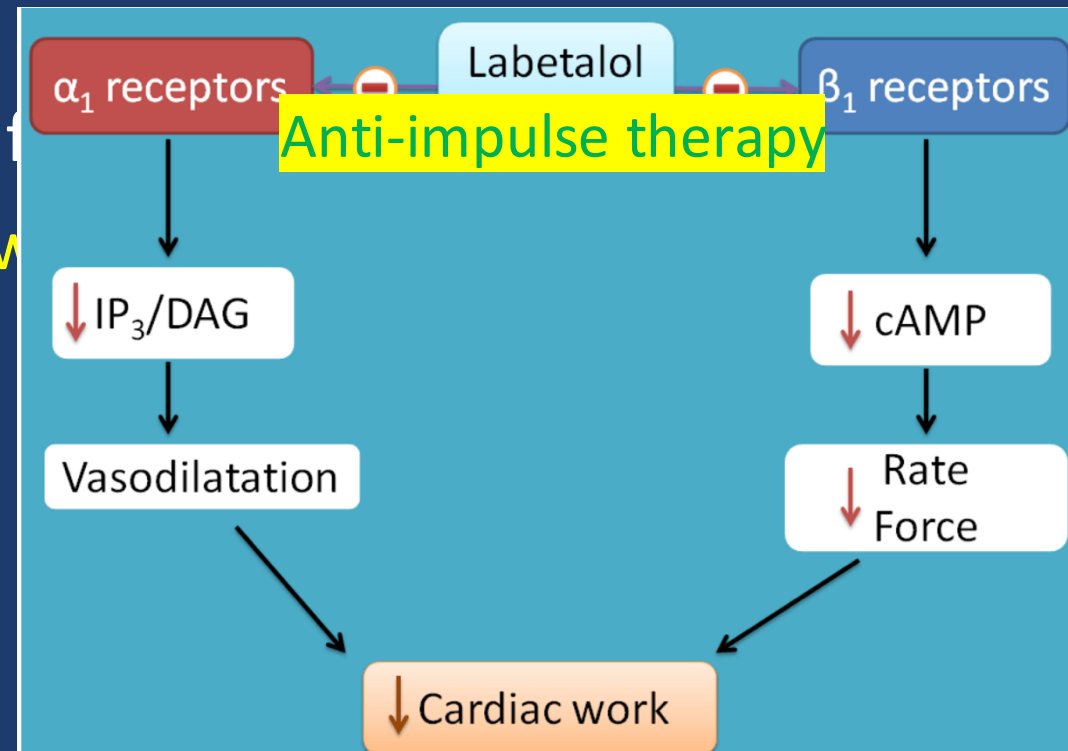
Nicardipine

- ❖ Calcium-channel blockers
- ❖ vasodilation and decreases peripheral resistance
- ❖ Half-life: 2-4 hr
- ❖ Concentration; 1:10, 1:5, 1:1
- ❖ Maximum rate = 15 mg/hr



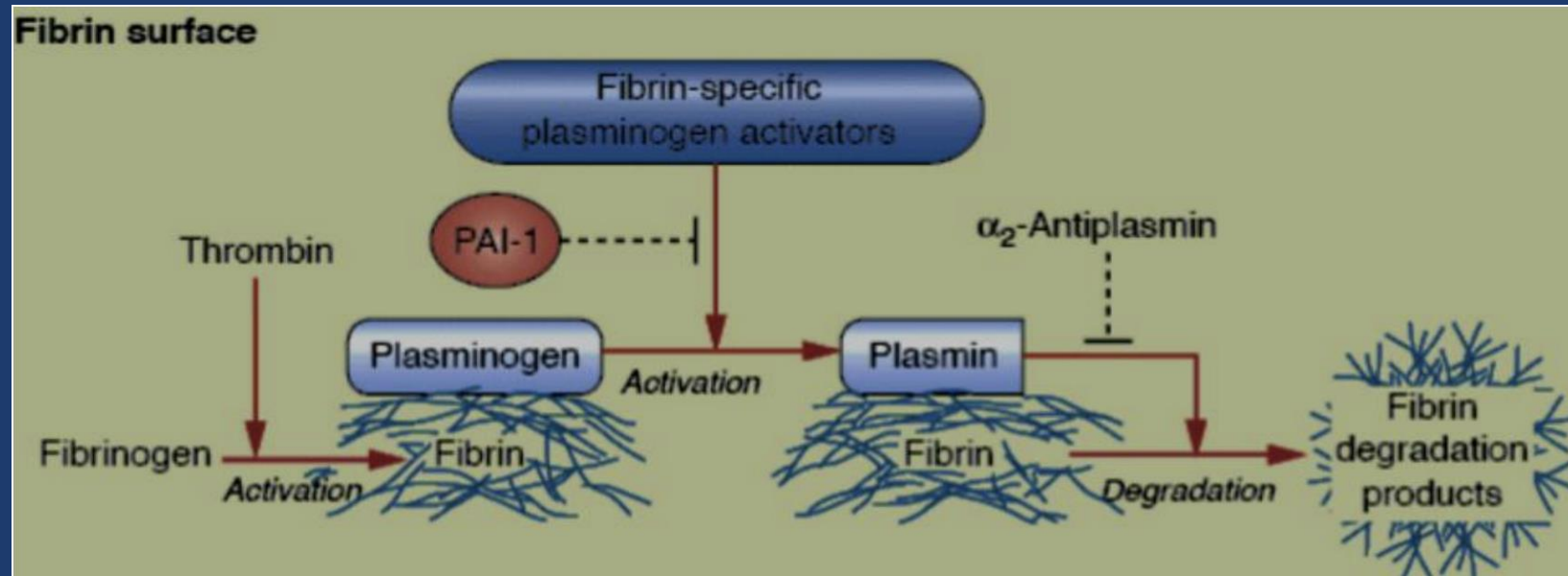
● Labetalol

- ❖ Beta-Blocker
- ❖ 20 mg IV over 2 minutes initially, then 40-80 mg IV q10min; total dose not to exceed 300 mg
- ❖ Alternative: 1-2 mg/min by continuous IV inf
- ❖ **Contraindication; Asthma or obstructive airway disease, severe bradycardia, 2°/3° heart block**



● rTPA

- ❖ Recombinant tissue plasminogen activator
- ❖ continuous, 0.5 to 1.0 mg/kg/h (40 mg maximum)
- ❖ Should monitor; fibrinogen > 200
- ❖ Beware fatal bleeding





rtPA

❖ Contraindication

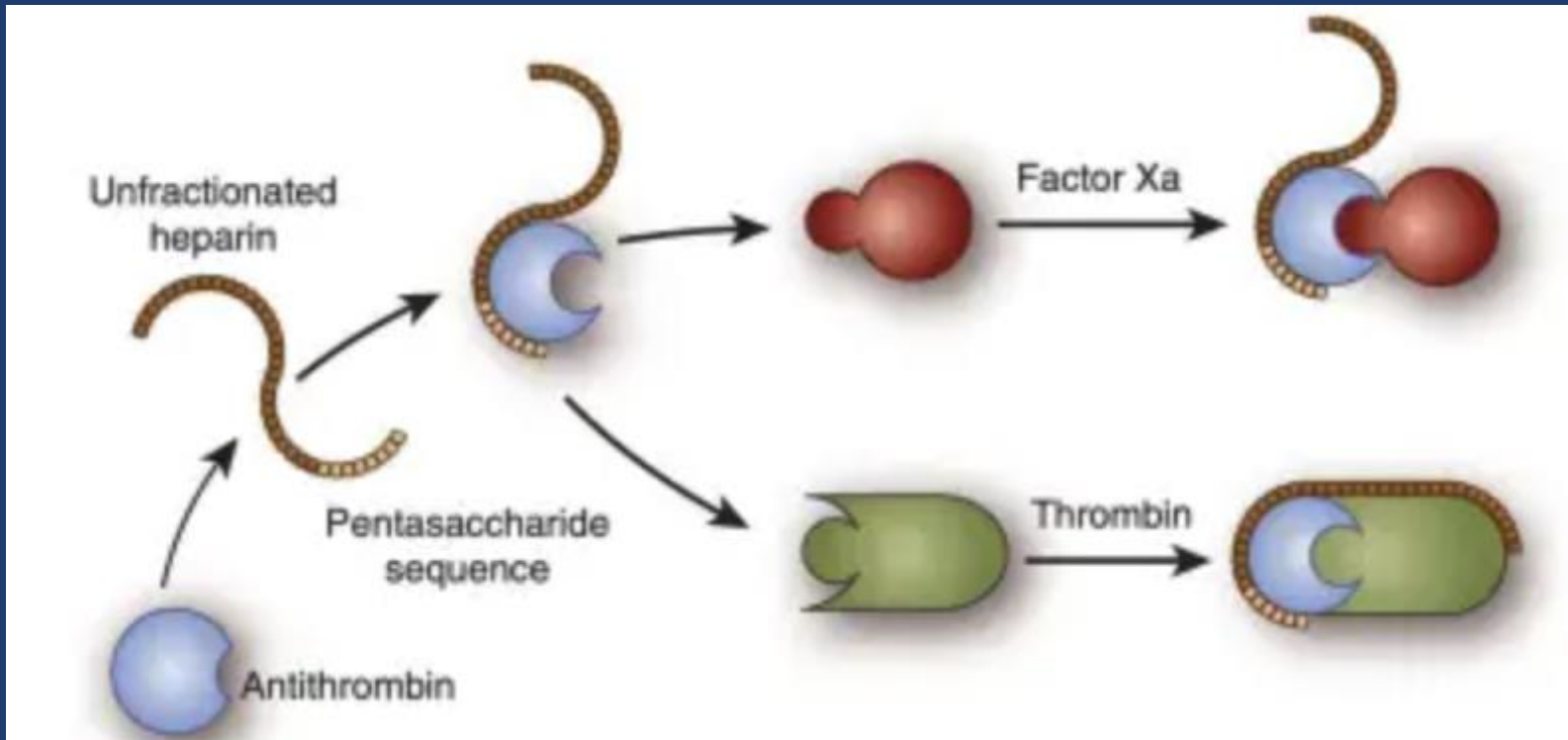
BOX 104.1

Contraindications to Pharmacologic Thrombolytic Agents

Major Contraindications

- Active bleeding disorder
- Gastrointestinal bleeding within 10 days
- Cerebrovascular accident within 6 months
- Intracranial or spinal surgery within 3 months
- Head injury within 3 months

● Unfractionated heparin



- ❖ Half life 1-2hr
- ❖ Keep aPTT ratio 1.5-2
- ❖ **Complication: HIT**



Case-based learning



PSU VASCULAR
PRINCE OF SONGKLA UNIVERSITY

THANK YOU

